Contract inference is a shared-context, multi-decision workload

A model can keep the same average accuracy while changing which contract decisions are right.

1 ContractNLI: one document, many structured judgments

Non-Disclosure Agreement (excerpt)

1 This Non-Disclosure Agreement ("Agreement") is made between the Disclosing
2 Party and the Receiving Party (each a "Party").
3 2. Confidential Information. The Receiving Party shall keep all Confidential
4 Information in strict confidence and shall not use such information except
5 for the purpose of evaluating a potential business relationship.
6 3. No Copying. The Receiving Party shall not copy, reproduce, or distribute
7 any Confidential Information, in whole or in part, by any means, without the
8 prior written consent of the Disclosing Party.
9 ...
10 6. Return or Destruction. Upon termination or expiration of this Agreement,
11 the Receiving Party shall promptly return to the Disclosing Party, or destroy,
12 all copies of the Confidential Information in its possession, including any
13 materials derived therefrom.

17 hypotheses about the contract (examples)

ID	Hypothesis	Label
nda-1	The Receiving Party must keep information confidential.	entailment
nda-2	The Receiving Party is allowed to copy the information for internal use.	contradiction
nda-3	The Receiving Party may share the information with its affiliates.	not_mentioned
nda-4	The agreement requires the return or destruction of materials after termination.	entailment
...	...	
nda-17	The Disclosing Party must provide technical support.	not_mentioned

Why this workload fits Jev

- One shared long document, many independent judgments.
- Each judgment is a bounded choice (entailment / contradiction / not_mentioned).
- No free-form generation is required.
- Jev supports shared state with multiple structured Choice judgments, avoiding free-form generation.

Architectural hypothesis: a decision-native model may be especially efficient for shared-document, multi-judgment workloads.

2 The same judgment can change with different request configurations

Fixed: the same contract, the same target hypothesis (nda-2), and the same gold label (contradiction).

Condition A: only the target hypothesis is shown

Prompt to the model (simplified)

Document: [full contract text]
Hypothesis (nda-2):
The Receiving Party is allowed
to copy the information
for internal use.
Task: Choose one label:
entailment / contradiction / not_mentioned.

Model: Qwen3.5-4B (3 repeated runs)

Run	Prediction	Correct?
1	contradiction	✓
2	contradiction	✓
3	contradiction	✓

3 / 3 correct

Only the request configuration changes (we add 16 additional non-target hypotheses as visible context).

Condition B: all 17 hypotheses shown (target: nda-2)

Prompt to the model (simplified)

Document: [full contract text]
Hypotheses:
nda-1: ...
nda-2: The Receiving Party is ...
nda-3: ... ... nda-17: ...
Task: Answer for nda-2 only.
Choose one label.

Model: Qwen3.5-4B (3 repeated runs)

Run	Prediction	Correct?
1	not_mentioned	✗
2	not_mentioned	✗
3	not_mentioned	✗

0 / 3 correct

Repeat-consistent, but configuration-sensitive.

Within each condition, the model is repeat-consistent. Yet its answer flips from consistently correct to consistently wrong when additional non-target hypotheses are shown. Agreement does not guarantee reliable correctness.

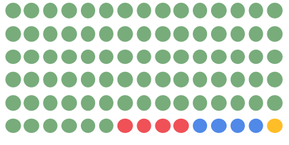
The same judgment can flip from consistently correct to consistently wrong.

3 Average accuracy can hide decision changes

Example: Qwen4B, comparing Condition A vs. B on 30 contracts (90 paired judgments).

Condition A	Condition B
Accuracy	Accuracy
75.6%	75.6%

Paired decision changes (n = 90)

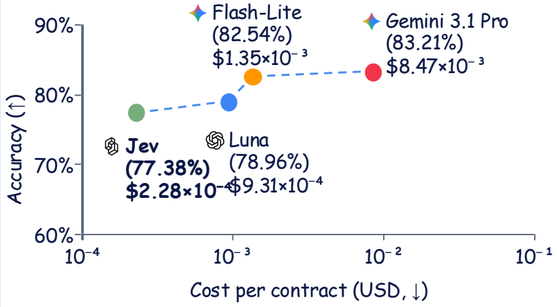


- Unchanged (81)
- Correct → Wrong (4)
- Wrong → Correct (4)
- Wrong → Other wrong (1)

Accuracy is identical, but 9 / 90 decisions changed.

Equal average accuracy can hide compensating corrections and regressions.

4 Model cost–accuracy trade-offs



Jev occupies the lowest-cost operating point in this comparison, while hosted LLMs reach higher baseline accuracy at higher cost.

Latency is also part of efficiency (e.g., Jev's baseline median latency is low, ~0.4 s).

5 Our evaluation considers four aspects

- **Cost** — How much does it cost to obtain the decisions?
- **Response time** — How quickly can the decisions be returned?
- **Average correctness** — How often are the decisions correct under a standard setting?
- **Persistent correctness** — Do the same decisions remain correct when the request configuration changes?

Evaluating contract inference requires efficiency, average correctness, and persistent correctness together.